Ayushman Jena

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Education

National Institute of Technology, Trichy, Tamil Nadu Coursework & Thesis Completion – Jan 2027
MS by Research, Instrumentation and Control Engineering *Degree Award – Aug 2027, CGPA: 8.75/10.0*

Veer Surendra Sai University of Technology, Burla, Odisha June 2024
B.Tech. Electrical Engineering *CGPA: 7.65/10.0*

Delhi Public School, Dhenkanal April 2020
Intermediate *Percentage: 96.2%*

Delhi Public School, Dhenkanal April 2018
Matriculation *Percentage: 93.6%*

Test Scores

IELTS Academic 2026
English Language Proficiency *Overall Band: 8.0*

Graduate Aptitude Test in Engineering 2024
Electrical Engineering *Qualified*

Skills

Skill-set: Deep Learning, Artificial Intelligence, Robotic design and development, Simulation and Modeling, Embedded PCB Design, CAD and FEM Analysis
Tools: Git/GitHub, MATLAB, VS Code/Jupyter Notebook, Arduino/CUBE/Esspressif IDE, TINA TI/LT SPICE, KICAD, Solidworks, COMSOL Multiphysics
Soft Skills: Project Management, Public Speaking, Crtical Thinking, Time management, Leadership

Experience

National Institute of Technology, Tiruchirappalli | *Senior Research Fellow* 18/10/2024 – Present
Currently heading research in the field of high frequency Focused Ultrasound sponsored by DRDO. Working on robotic scanning setups, control implementation, sensor data processing and machine learning applications in signal processing and system modelling

Shyam Metallics and Energy Limited, Odisha | *Graduate Engineer Trainee* 01/07/2024 – 01/09/2024
Worked at Benefication Plant and Pellet Plant for electrical maintenance and control of 33 Kv and 11 Kv Induction Motors, transformers.

Defense Research Development Organization, ITR Chandipur | *Summer Trainee* 01/06/2023 – 30/06/2023
Worked at Central Data Processing division. Studied ballistic motion and sample data generation in MATLAB, noise and error, estimators, and filters. Explored the feedback control system of onboard flight computers of missiles. Implemented Recursive Kalman filter on noisy ballistic motion data simulating flight motion.

Veer Surendra Sai Space Innovation Center(VSSSIC) | *Student Coordinator* 01/2023 – 01/2024
Worked on VSLV, a mega project recognized by ISRO as per the MoU between VSSUT, Burla and ISRO. Responsible for flight controller development, estimation, PCB design, trajectory simulations and overall project develpment.

My Equation Pvt.Ltd. | *Intern — Electrical Domain Head* 05/2021–07/2022
Attended and mentored in workshops on EV, Industrial Automation, and Robotics. Worked on Uni-cycle-based campus transport research and design of electrical-electronic subsystem.

Courses/Certifications

- Computer Science & Robotics - Learn by Doing! Linux to ROS 2 Udemy
- Physical AI Foundations: The Math Behind VLMs, VLAs and WAM Udemy
- Machine Learning for High Energy Physics Preschool, (ML4HEP 2026) by TIFR Mumbai
- Computational Physics: Scientific Programming with Python Udemy
- Embedded C Programming, Embedded System Design and IoT, By PANTECH SOLUTIONS
- Application of Machine Learning Techniques in Sustainable Technologies, (AMLST 2024) by NIT Rourkela
- Python Essentials 1, By CISCO NETWORKING ACADEMY
- CADET at 3 O CTC NCC, VSSUT Burla, B Certificate, C Certificate holder with grade A.

Achievements

- National Finalist at CANSAT Student India Competition 2024 by INSPACe & ASI at Ahmedabad. Proud Leader of the only team from Odisha.
- Secured 3rd position at Odisha Technological Conclave Model Exhibition for project VSLV, 15th Sept 2023
- Secured 2nd position at World Space Week by SDSC ISRO, Innovative Design Criteria for RF Harvesting CubeSat.
- Exhibit at 3rd International Conference on Ranged Technologies (ICORT) held at ITR DRDO Chandipur 2023.

Projects

Reconstruction and Feature Extraction of Focused Ultrasound Fields using CNN | *COMSOL Multiphysics, VS Code*
- Work in Progress: 2D and 3D reconstruction of focused ultrasound from COMSOL simulation and experimental hydrophone data for acoustic feature extracion using CNN.

PINNs based Piezo-Bio-Acoustic Simulation for tFUS | *System Modeling, PDEs, Optimization, VS Code*
- Work in Progress: Development of an accurate simulation model of generation, propagation and effect of focused ultrasound in respect with the human head.

Transcranial Focused Ultrasound Stimulation Device | *Solidworks, COMSOL Multiphysics, Python, MATLAB*
- Design and development of TFUS device for cognitive enhancement in military scenario is a DRDO LSRB funded project which I joined as a Junior Research Fellow.

Modified Ultrasonic C Scan — Gantry Robotic Manipulator Setup | *VS Code, Embedded C, Python, NMPC*
- Developed a 3 DOF inverted robotic manipulator mounted on a 2 axis motion setup elevated 0.5m above the surface with real time nonlinear model predictive controller.

VSSUT Satellite Launch Vehicle and CANSAT | *Embedded C, Python, Sensor Fusion, PCB Design, MATLAB*
- Sounding rocket project developed for studying the Hirakud dam and the Mahanadi reservoir for environmental implications funded by ISRO, VSSUT. CAN-sized satellite made to track trajectory, control mid-flight orientation and produce necessary data with a secondary recovery system.

Patents/Publications

Research Publications
- Design and Experimentation of Nonlinear Model Predictive Control for Gantry Robotic Manipulator, IEEE Xplore, I5CPS, 2026
- Reliability analysis of heavy earth machinery, for Predictive Scheduling using the Kolmogorov-Smirnov Probability Test and Convolutional Neural Network, IJARIIT Vol-7, Issue 6, 2021

Conference Presentations (Unpublished Works)
- Optimization of Bezier curve using Genetic algorithm for design of Bell Shape Nozzle Contour, ICRAMERD 2024, ITER Bhubaneswar, Best Session Presentation Award
- Design of a Single Element Spherical Cap Transducer for Transcranial Focused Ultrasound Stimulation, MBCC 2025, IIT Mandi